

Table 1: 第一級繞射

n_i	繞射角 θ_{obs} ($^\circ$)	λ_{obs} (nm)	λ_{ref} (nm)	Error	Rydberg constant (nm^{-1})
3	23.47, 23.30	661.5	656.3	0.79%	0.0108849
4	17.13, 17.00	489.1	486.1	0.63%	0.0109035
5	15.28, 15.17	437.7	434	0.85%	0.0108798
6	14.42, 14.33	413.8	410.2	0.87%	0.0108754

Table 2: 第二級繞射

n_i	繞射角 θ_{obs} ($^\circ$)	λ_{obs} (nm)	λ_{ref} (nm)	Error	Rydberg constant (nm^{-1})
3	52.42, 53.07	663.3	656.3	1.06%	0.0108556
4	36.00, 36.02	489.9	486.1	0.79%	0.0108862
5	31.67, 31.73	437.9	434	0.90%	0.0108746
6	29.73	413.3	410.2	0.76%	0.0108879

Table 3: 第三級繞射

n_i	繞射角 θ_{obs} ($^\circ$)	λ_{obs} (nm)	λ_{ref} (nm)	Error	Rydberg constant (nm^{-1})
4	61.55	488.5	486.1	0.49%	0.0109186
5	51.88	437.1	434	0.71%	0.0108947

Table 4: fitting result for $m = 1$

Quantity	Value	Error
$-s$ (m^{-1})	$(1.0861 \pm 0.0044) \times 10^7$	-0.98%
$4b$ (m^{-1})	$(1.0879 \pm 0.0012) \times 10^7$	-0.81%
x-intercept	0.2504 ± 0.0013	0.17%

Table 5: fitting result for $m = 2$

Quantity	Value	Error
$-s \text{ (m}^{-1}\text{)}$	$(1.0930 \pm 0.0028) \times 10^7$	-0.34%
$4b \text{ (m}^{-1}\text{)}$	$(1.08905 \pm 0.00077) \times 10^7$	-0.70%
x-intercept	0.24909 ± 0.00080	-0.36%

Table 6: 物理常數的計算結果

Quantity	Value
$R_{\text{est}} \text{ (m}^{-1}\text{)}$	$(1.08861 \pm 0.00055) \times 10^7$
$R_{\text{theo}} = \frac{\mu e^4}{8 \varepsilon_0^2 h_{\text{ref}}^3 c} \text{ (m}^{-1}\text{)}$	$1.097e + 07$
Error	-0.74%
$h_{\text{est}} \text{ (J s)}$	$(6.6426 \pm 0.0011) \times 10^{-34}$
$h_{\text{ref}} \text{ (J s)}$	$6.626e - 34$
Error	0.25%

Table 7: 不同譜線對應的折射率

n_i	最小偏折角 D	折射率 n
3	36.93	1.497
4	37.62	1.505
5	38.02	1.51
6	38.6	1.516